

MIXED PRECISION DNNs: ALL YOU NEED IS A GOOD PARAMETRIZATION

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ABSTRACT

Efficient deep neural network (DNN) inference on mobile or embedded devices typically involves quantization of the network parameters and activations. In particular, mixed precision networks achieve better performance than networks with homogeneous bitwidth for the same size constraint. Since choosing the optimal bitwidths is not straight forward, training methods, which can learn them, are desirable. Differentiable quantization with straight-through gradients allows to learn the quantizer’s parameters using gradient methods. We show that a suited parametrization of the quantizer is the key to achieve a stable training and a good final performance. Specifically, we propose to parametrize the quantizer with the step size and dynamic range. The bitwidth can then be inferred from them. Other parametrizations, which explicitly use the bitwidth, consistently perform worse. We confirm our findings with experiments on CIFAR-10 and ImageNet and we obtain mixed precision DNNs with learned quantization parameters, achieving state-of-the-art performance.

1 INTRODUCTION

Quantized DNNs apply quantizers $Q : \mathbb{R} \rightarrow \{q_1, \dots, q_I\}$ to discretize the weights and/or activations of a DNN (Han et al., 2015; Zhou et al., 2017; Li et al., 2016; Liu & Mattina, 2019; Cardinaux et al., 2018; Jain et al., 2019; Bai et al., 2018). They require considerably less memory and have a lower computational complexity, since discretized values $\{q_1, \dots, q_I\}$ can be stored, multiplied and accumulated efficiently. This is particularly relevant for inference on mobile or embedded devices with limited computational power.

However, gradient based training of quantized DNNs is difficult, as the gradient of a quantization function vanishes almost everywhere, i.e., backpropagation through a quantized DNN almost always returns a zero gradient. Different solutions to this problem have been proposed in the literature:

A first possibility is to use DNNs with stochastic weights from a categorical distribution and to optimize the *evidence lower bound* (ELBO) to obtain an estimate of the posterior distribution of the weights. As proposed in (Jang et al., 2016; Maddison et al., 2016; Louizos et al., 2019), the categorical distribution can be relaxed to a concrete distribution – a smoothed approximation of the categorical distribution – such that the ELBO becomes differentiable under reparametrization.

A second possibility is to use the *straight through estimator* (STE) (Bengio et al., 2013). STE allows the gradients to be backpropagated through the quantizers and, thus, the network weights can be adapted with standard gradient descent (Hubara et al., 2016). Compared to STE based methods, stochastic methods suffer from large gradient variance, which makes training of large quantized DNNs difficult. Therefore, STE based methods are more popular in practice.

More recent research (Jain et al., 2019; Esser et al., 2019; Wang et al., 2018; Elthakeb et al., 2018) focuses on methods which can also learn the optimal quantization parameters, e.g., the stepsize, dynamic range and bitwidth, in parallel to the network weights. This is a promising approach as DNNs with learned quantization parameters almost always outperform DNNs with handcrafted ones.

*Equal contribution.

The source code can be found at <https://github.com/sony/ai-research-code>.